MTH208

Examination - January Semester 2025

Advanced Linear Algebra

Tuesday, 22 April 2025 **01:00 pm - 03:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **SIX (6)** questions and comprises **SEVEN (7)** printed pages (including cover page).

2. You must answer **ALL** questions.

3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.

4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1

Let

$$A = \begin{pmatrix} 1 & -17 & -6 & 19 \\ 1 & 31 & 11 & -32 \\ 1 & -21 & -7 & 23 \\ 1 & 19 & 7 & -19 \end{pmatrix}$$

and let $V = \mathbb{C}^4$. Define a linear operator $T : V \longrightarrow V, T(x) = Ax$.

Question 1a

Show that A has an eigenvalue $1 + i$ with corresponding eigenvector

$$\begin{pmatrix} -\frac{21}{26} - \frac{1}{26}i \\ \frac{41}{26} - \frac{3}{26}i \\ -\frac{17}{13} + \frac{6}{13}i \\ 1 \end{pmatrix}.$$

Prove also that A has an eigenvalue 2 and find a corresponding eigenvector.

(6 marks)

Question 1b

Given that the characteristic polynomial of A is $(x-2)^2(x^2-2x+2)$, evaluate $(A-2I)(A^2-2A+2I)$ and conclude that the minimal polynomial of A is also $(x-2)^2(x^2-2x+2)$. Using the results of Question 1a or otherwise, solve for the Jordan canonical form of A.

(7 marks)

Question 1c

Determine T -invariant subspaces W_1 and W_2 of V such that $V = W_1 \oplus W_2$ where $\dim W_1 = 2 = \dim W_2$. Give a basis for each subspace.

(7 marks)

Question 2

Let V denote the vector space of polynomials of degree at most 2 over $\mathbb{R}$. Let

$$\mathfrak{B} = \{(1-x)^2, 2x(1-x), x^2\}$$

be a basis for V, and let $\mathfrak{B}^* = \{f_1, f_2, f_3\}$ be the dual basis of $\mathfrak{B}$.

Question 2a

Compute the matrix $D = (d_{ij})_{1 \leq i,j \leq 3}$ such that

$$\begin{aligned} f_1(a + bx + cx^2) &= d_{11}a + d_{12}b + d_{13}c \\ f_2(a + bx + cx^2) &= d_{21}a + d_{22}b + d_{23}c \\ f_3(a + bx + cx^2) &= d_{31}a + d_{32}b + d_{33}c \end{aligned}$$

for every polynomial $a + bx + cx^2 \in V$.

(5 marks)

Question 2b

Let W be the subspace of V spanned by the polynomial $1 + x + x^2$, and let W^0 be the subspace of V^* that annihilates W. Find a basis for W^0, expressing each basis vector as a linear combination of f_1, f_2, f_3.

(8 marks)

Question 2c

Let h_1, h_2, h_2 be linear functionals in V^* satisfying

$$h_1(p(x)) = p(-2), h_2(p(x)) = p(-1), h_3(p(x)) = p(1)$$

for every polynomial $p(x) \in V$. Prove that $\mathfrak{D}^* = \{h_1, h_2, h_3\}$ is a basis for V^*, and find a basis $\mathfrak{D}$ for V such that $\mathfrak{D}^*$ is the dual basis of $\mathfrak{D}$.

(7 marks)

Question 3

Suppose $\mathfrak{B} = \{w_1, w_2, w_3\}$ is a linearly independent set of vectors in a real inner product space V such that

$$\langle w_1, w_1 \rangle = 8, \langle w_1, w_2 \rangle = 4, \langle w_1, w_3 \rangle = 4,$$
$$\langle w_2, w_2 \rangle = 16, \langle w_2, w_3 \rangle = 8,$$

where $\langle \cdot, \cdot \rangle$ denotes the inner product on V. Let $W = span(\mathfrak{B})$ and define a linear operator $T : W \longrightarrow W$ by

$$T(w_1) = w_1 - 2w_2 + w_3,$$
$$T(w_2) = w_2 + 3w_3,$$
$$T(w_3) = w_1 - w_3.$$

Question 3a

Use the Gram-Schmidt process to determine an orthogonal basis $\mathfrak{C} = \{v_1, v_2, v_3\}$ for W, expressing each of the vectors v_1, v_2, v_3 as a linear combination of w_1, w_2, w_3.

(6 marks)

Question 3b

Compute $[T]_{\mathfrak{C}}$, the matrix representation of T relative to the orthogonal basis $\mathfrak{C}$ derived in Question 3a.

(4 marks)

Question 4

Let

$$A = \begin{pmatrix} 10 & 15 \\ -15 & 6 \end{pmatrix}$$

and let $B = AA^*$, a normal matrix.

Question 4a

Prove that the eigenvalues of B are 225 and 361, and illustrate the property of normal operators by constructing a unitary matrix Q and a diagonal matrix D such that $Q^*BQ = D$.

(8 marks)

Question 4b

Find a singular value decomposition of A.

(5 marks)

Question 4c

Suppose C is an $n \times n$ matrix with entries in $\mathbb{C}$. Let $U\Sigma V^*$ be the singular value decomposition of C, where U are V are $n \times n$ unitary matrices and Σ is an $n \times n$ matrix with $rank(C)$ leading positive diagonal entries and zero in other entries. Give brief explanations why UV^* is unitary and $V\Sigma V^*$ is positive semidefinite. Hence derive the polar decomposition of C; that is, show that $C = U'P$ for a unitary matrix U' and a positive semidefinite matrix P.

(3 marks)

Question 4d

Using the results of Questions 4b and 4c, or otherwise, find a polar decomposition of A.

(4 marks)

Question 5

Let

$$A = \begin{pmatrix} 0 & 2 & -4 \\ 2 & 4 & 3 \\ -4 & 3 & -4 \end{pmatrix}$$

be the matrix representation of a symmetric bilinear form f on $\mathbb{R}^3$ relative to the standard basis. Compute a diagonal representation of f by finding an invertible matrix Q such that QAQ^T is a diagonal matrix, and determine the rank and signature of f. Determine whether A is positive definite.

(15 marks)

Question 6a

Let

$$x_1 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}, x_2 = \begin{pmatrix} 3 \\ 2 \end{pmatrix}, x_3 = \begin{pmatrix} 5 \\ 6 \end{pmatrix}, x_4 = \begin{pmatrix} 6 \\ 5 \end{pmatrix}$$

be four vectors in $\mathbb{R}^2$. Let X be the 4×2 matrix whose i-th row is $x_i^\top$ and let u be a **unit** eigenvector of $X^\top X$ corresponding to the largest eigenvalue of $X^\top X$. (Note that $X^\top X$ is positive-definite.) For each $i, 1 \leq i \leq 4,$ the scalar $u^\top x_i$ may be regarded as a one-dimensional representation of x_i ; this one-dimensional representation can then be multiplied by u to approximately reconstruct x_i. Compute the reconstruction error

$$\sum_{i=1}^{4} \|x_i - uu^\top x_i\|_2^2,$$

where $\|\cdot\|_2$ denotes the Euclidean norm on $\mathbb{R}^2$.

(7 marks)

Question 6b

Suppose

$$y_1 = \begin{bmatrix} 2 \\ 3 \\ 5 \\ 6 \end{bmatrix} \text{ and } y_2 = \begin{bmatrix} 3 \\ 2 \\ 6 \\ 5 \end{bmatrix}.$$

Let Y be the 2×4 matrix whose i-th row is $y_i^\top$; note that $Y = X^\top$, where X is defined as in Question 6a. Based on the eigenvalues of $X^\top X$, determine the two largest eigenvalues of $Y^\top Y$. Find a 4×2 matrix U whose first column is a **unit** eigenvector of $Y^\top Y$ corresponding to the largest eigenvalue of $Y^\top Y$ and whose second column is a **unit** eigenvector of $Y^\top Y$ corresponding to the second largest eigenvalue of $Y^\top Y$, and hence compute the two-dimensional representations $U^\top y_1$ and $U^\top y_2$ of y_1 and y_2 respectively. Give your answers in exact form, including a surd if necessary.

(8 marks)

----- END OF PAPER -----